

Gravity Marketing & Communications

DEVELOPMENT PLAN

Cross-Platform Digital Ticketing & Event Discovery App (Ethiopia)

Project Codename: EtTicket

Platforms: Android, iOS, Web (Admin & Organizer)

Target Market: Ethiopia (Addis Ababa → Nationwide)

Launch Goal: MVP in 6 months, scalable for 3+ years

1. PRODUCT GOAL (FOR DEVELOPERS)

Build a **secure, scalable, Ethiopia-localized digital ticketing platform** that allows:

- Users to discover events & movies
- Buy tickets using local payments
- Receive QR/OTP digital tickets
- Validate tickets on-site (online & offline)
- Allow organizers to manage events and sales
- Allow admins to control commissions, payouts, fraud, and analytics

Comparable product: **BookMyShow (India)**

Localization focus: **Ethiopia-first**

2. CORE MODULES (SYSTEM BREAKDOWN)

A. User Mobile App (B2C)

- Android (Priority)
- iOS
- Built with **single cross-platform codebase**

B. Organizer Web Dashboard (B2B)

- Event & ticket management
- Sales & analytics

C. Admin Web Dashboard

- Platform control & monitoring

D. Backend & APIs

- Business logic
- Payments
- Ticketing
- Security
- Analytics

E. Ticket Validation System

- QR scanning
 - Offline support
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3. TECHNOLOGY STACK (MANDATORY)

Frontend

- **Flutter** (Android & iOS)
- **React.js** (Web dashboards)

Backend

- **Node.js + Express** (*or Django REST — team chooses one*)
- RESTful APIs
- JWT authentication

Database

- **PostgreSQL** – main database
- **Redis** – seat locking, caching, OTPs

Payments (Phase 1)

- Telebirr
- CBE Birr
- Amole
- Bank transfer reference
- Cash-at-venue (QR validation only)

Infrastructure

- AWS or GCP
 - Dockerized services
 - Object storage for images (S3/GCS)
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4. FUNCTIONAL REQUIREMENTS (DETAILED)

4.1 User App (B2C)

Authentication

- Phone number registration
- OTP verification
- Optional email
- Language selection (Amharic / English)

Event Discovery

- Browse by:
 - City
 - Category
 - Date
- Search events
- Featured & promoted events

Event Details

- Event description
- Venue & location
- Date & time
- Ticket types (Regular, VIP, etc.)
- Seating map (if applicable)
- Availability status

Ticket Booking

- Select seat / zone / quantity
- Real-time seat locking (Redis)
- Price breakdown:
 - Ticket price
 - Commission
 - Convenience fee
- Promo code support

Payments

- Redirect or in-app payment
- Backend verification (no client-side trust)
- Payment retry handling

Ticket Generation

- Unique Ticket ID
- Encrypted QR code
- Backup OTP code
- Offline storage in app
- SMS confirmation (optional)

My Tickets

- Upcoming tickets
- Past tickets
- Ticket status (Valid / Used / Cancelled)

Notifications

- Booking confirmation
 - Event reminders
 - Event updates/cancellations
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4.2 Organizer Dashboard (B2B)

Organizer Account

- Registration
- Admin approval required
- Bank/mobile money details

Event Management

- Create & edit events
- Upload images
- Set ticket types & prices
- Set capacity or seat map
- Define refund rules

Ticket Sales

- Real-time sales dashboard
- Tickets sold per type
- Revenue summary

Promotions

- Promo codes
- Discounts
- Featured listings (admin-approved)

Reports

- CSV export
- Sales by date
- Attendance report

Ticket Validation

- QR scanner (web or mobile)
 - Offline scan mode
 - Sync scans when online
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4.3 Admin Dashboard

Platform Control

- Approve organizers & events
- Set commission % per organizer/event
- Set convenience fees

Financials

- GMV tracking
- Platform revenue
- Organizer payouts
- Settlement status

Fraud & Security

- Duplicate QR detection
- Suspicious activity flags
- Manual ticket invalidation

Analytics

- Tickets sold (daily/monthly)
- Revenue by category & city
- Active users

Content Management

- Featured events
 - Homepage banners
 - Categories & cities
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5. TICKET VALIDATION LOGIC (CRITICAL)

Ticket Structure

- Ticket ID (UUID)
- Event ID
- User ID
- Encrypted QR payload
- OTP fallback

Validation Rules

- QR can be scanned **once**
- Status changes:
 - VALID → USED
- Offline scans stored locally
- Sync & reconcile later

Edge Cases

- Duplicate scan → Reject
 - Event cancelled → Auto invalidate
 - Manual override (admin only)
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6. SECURITY REQUIREMENTS

- JWT auth with refresh tokens
 - Role-based access (User / Organizer / Admin)
 - Encrypted QR payload
 - HTTPS only
 - Rate limiting on OTP & payment APIs
 - No payment success from frontend without backend verification
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7. NON-FUNCTIONAL REQUIREMENTS

- Offline QR scanning support
- Low bandwidth optimization
- Fast load on Android low-end devices
- Scalable to 1.5M+ tickets/year
- Logs & monitoring enabled

8. DEVELOPMENT PHASES & TIMELINE (2 MONTHS)

Phase 1 – Planning & Architecture (Month 1)

Deliverables

- Final system architecture
 - Database schema
 - API specs
 - Wireframes (approved)
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Phase 2 – UI/UX Design (Month 1)

Deliverables

- Flutter UI designs
 - Web dashboard designs
 - User flow diagrams
 - Design handoff (Figma)
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Phase 3 – Core Development (Months 2)

Deliverables

- User app (booking + tickets)
 - Organizer dashboard
 - Admin dashboard
 - Payment integrations
 - QR ticket system
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Phase 4 – Testing & Pilot (Month 2)

Deliverables

- QA testing
- Load testing
- Payment testing
- Pilot with 1–2 cinemas/events

Phase 5 – Launch (Month 6)

Deliverables

- Play Store release
 - App Store submission
 - Production deployment
 - Monitoring setup
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9. MVP ACCEPTANCE CRITERIA (GO / NO-GO)

The MVP is **approved** if:

- ✓ Users can buy tickets end-to-end
 - ✓ Payments verify correctly
 - ✓ QR tickets scan once only
 - ✓ Organizers see sales in real time
 - ✓ Admin can control commissions & payouts
 - ✓ App works on low-end Android phones
 - ✓ Offline QR scanning works
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10. POST-LAUNCH (PHASE 2 FEATURES – OPTIONAL)

- Loyalty points
 - Wallet balance
 - Diaspora card payments
 - Seat map designer
 - White-label organizer pages
 - Event live streaming
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